

EE-482: Network Security and Cryptography

LECTURE NO 19: *Virus Structure*

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21. Virus Structure (1/10)

- ↳ Virus can be prepended or postpended to an executable program or it can be embedded in source program.
- ↳ Infected program, when invoked will first execute the virus code & then execute the original code of the program.

21. Virus Structure (2/10)

Example: Virus V, prepended to infected program

program V :=

```
{ goto main: // Jump to main virus program  
  1 2 3 4 5 6 7; // marker to determine if prog. infected
```

subroutine infect-executable :=

seeks uninfected
executable files
& infects them
PROPAGATION

```
{ loop:  
  file := get-random-executable-file;  
  if (first-line-of-file = 1 2 3 4 5 6 7)  
    then goto loop  
  else prepend V to file; }
```

subroutine do-damage :=

```
{ whatever damage is to be done }
```

triggers only
under certain
conditions

```
{ subroutine trigger-pulled :=  
  { return true if some condition holds }
```

21. Virus Structure (3/10)

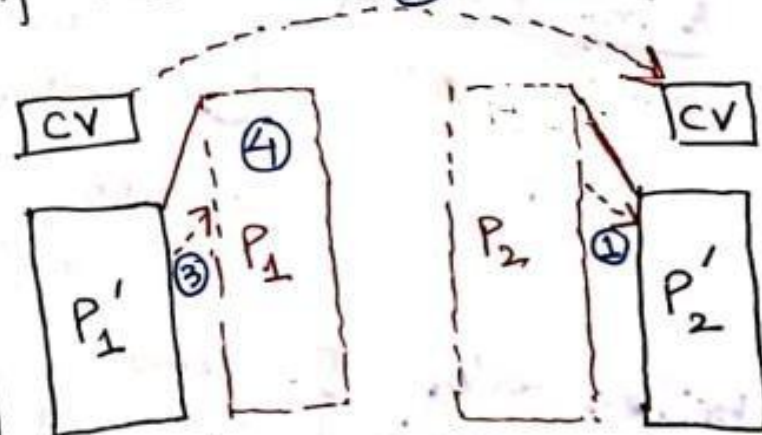
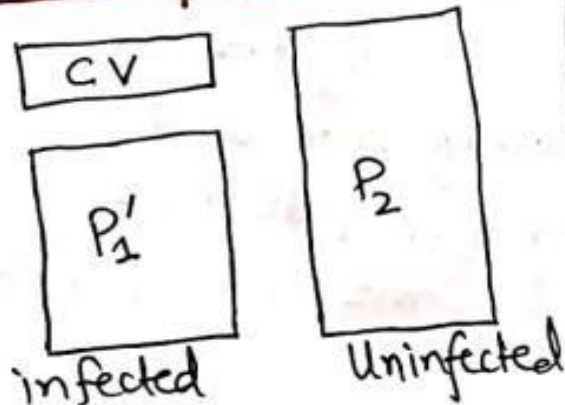
```
main: main program :=  
    { infect-executable;  
      if trigger-pulled;  
        then do-damage;  
        goto next; } // transfer control to original program  
next: // actual program executed  
}
```

21. Virus Structure (4/10)

Detection

- ↳ Prepend virus is easy to detect because infected version of program is longer than uninfected one.
- ↳ Can run antivirus check on the size of the file.
- ↳ To prevent detection by antivirus, compress the executable file so that both the infected & uninfected versions are of identical length.
- ↳ Compressed by the size of the virus. ②

A Compression Virus:



21. Virus Structure (5/10)

```
Program CV :=  
  { goto main;  
    01234567;  
    subroutine infect-executable :=  
      { loop:  
        file := get-random-executable-file;  
        if (first-line-of-file = 01234567)  
          then goto loop  
        else (1) compress file;  
              (2) prepend CV to file;  
main: main-program :=  
  { if ask-permission then infect executable;  
    (3) uncompress-rest-of-file;  
    (4) run-uncompressed-file; }
```

21. Virus Structure (6/10)

- Program P_1 is infected with virus CV.
- When program is invoked, control passes to virus which performs following functions:
 - (1) For each uninfected file P_2 , that is found, virus first compresses that file to produce P_2' which is shorter than original program by the size of virus.
 - (2) A copy of virus is prepended to the compressed program.
 - (3) compressed version of original infected program P_1' is uncompressed.

21. Virus Structure (7/10)

(4) Uncompressed original program is executed

↳ Once a virus gains entry to a system by infecting a single program, it is in a position to infect some or all executable files on that system.

↳ Viral infection can be completely prevented by preventing the virus from gaining entry in the first place.

↳ Prevent is very difficult.

21. Virus Structure (8/10)

Antivirus Approaches:

Ideal Approach: Prevention (do not allow a virus to get into system in the first place)

Next best approach:

- * Detection: Once infection has occurred, determine that it has occurred and locate the virus.
- * Identification: Once detection achieved, identify the specific virus that has infected a program.
- * Removal: Once identified, remove all traces of virus from infected program & restore it to its original state.

21. Virus Structure (9/10)

Types of Viruses: 5 main types

(1) Parasitic Virus:

- ↳ Traditional & most common type
- ↳ Attaches itself to executable files and replicates when infected program is executed by finding other files to infect.

(2) Memory Residential Virus:

- ↳ lodges/resides in main memory (RAM) as part of a resident program.
- ↳ Virus infects every program that executes

(3) Boot Sector Virus:

- ↳ Infects a master boot record or boot record.
- ↳ Spreads when system is booted.

21. Virus Structure (10/10)

(4) Stealth Virus:

↳ A form of virus designed to hide itself from detection by antivirus software.

Example: A virus that uses compression so that infected program is exactly same length as uninfected version.

(5) Polymorphic Virus:

↳ A virus that mutates with every infection.

↳ Creates copies during replication that are functionally equivalent but have distinctly different bit patterns.

↳ Purpose is to defeat antivirus programs which scan for signature of the virus.

↳ Signature will vary with each copy.

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*